

FUNDAMENTALS OF MATHEMATICS

Course Information

Degree: UG

Program: BCA

Course Name: FUNDAMENTALS OF MATHEMATICS

Course Credits: 3

Semester: I

Course Category:

Type of Course: Lecture

New Course: NO

Learning Hours

Activity	Hours per Week
Lecture	3
Tutorial	0
Practical	0
Experiential Learning	0

Course Contribution

Category	Percentage Contribution
Employment Enhancement	60%
Entrepreneurship Skills	30%
Skill Development	10%

Course Relevance

Level	Percentage Relevance
Global	40%
National	30%
Regional	20%
Local	10%

Course Outcomes (COs)

CO#	Course Outcome	Bloom's Taxonomy Level
CO1	Identify and Analyze statements and prove the concepts of Mathematical logics.	BL-2
CO2	Explain the concept of Matrices	BL-3
CO3	Discuss the concept of the system of equations.	BL-5
CO4	Analyze Understand and Explain set theory	BL-3
CO5	Identify and Implement the concept of Permutation and combination	BL-4

Module	Course Content	Tools Used / Assessment and Activity	Bloom's Level	Mapping COs	Percentage change in Module (put Nil if there is no change or it is a new course write NEW)
Module 1: [Mathematical Logic]	Proposition		BL-2	1	OLD
	compound proposition		BL-3	1	OLD
	logical connectives – problems		BL-2	1	OLD
	laws of logics		BL-2	1	OLD
	problems on laws of logic		BL-2	1	OLD
	Open sentences		BL-2	1	OLD
	Truth sets		BL-2	1	OLD
	Connectives involving quantifiers		BL-2	1	OLD
	Logical implications.				OLD

Module 2: [Matrix Theory]	Types of matrices		BL-4	2	OLD
	algebra of matrices–addition		BL-3	2	OLD
	subtraction		BL-3	2	OLD
	multiplication of matrices		BL-2	2	OLD
	determinant of a matrix		BL-3	2	OLD
	symmetric and skew-symmetric matrices		BL-2	2	OLD
	orthogonal matrix		BL-3	2	OLD
	rank of a matrix		BL-3	2	OLD
	inverse of a matrix		BL-4	2	OLD
Module 3: [System of linear equations]	Linear equations introduction		BL-2	3	OLD
	Homogeneous linear equations,		BL-3	3	OLD
	Working rule of Homogeneous linear equations,		BL-3	3	OLD
	Non – homogenous equations.		BL-3	3	OLD
	Working rule of Non – homogenous equations.		BL-3	3	OLD
	Trivial equations and problems		BL-3	3	OLD
	non trivial equations and problems		BL-4	3	OLD
Module 4: [Set theory, Relations and functions]	Set Theory : Fundamentals of set operations		BL-2	4	OLD
	Subsets		BL-3	4	OLD
	Counting and Venn Diagrams		BL-3	4	OLD
	Relations		BL-2	4	OLD
	Functions: Plain and one to one		BL-3	4	OLD
	Onto Functions		BL-3	4	OLD
	Function Composition		BL-3	4	OLD

	Inverse Functions		BL-3	4	OLD
Module 5: [Permutat ions and Combinati ons]	Permutations and Combinations :Introduction		BL-3	5	OLD
	The rule of sum and product		BL-3	5	OLD
	Permutations problems		BL-3	5	OLD
	Permutations with Repetitions		BL-3	5	OLD
	Permutations of Objects not all Different		BL-3	5	OLD
	Combinations problems		BL-3	5	OLD
	Combinations with Repetition		BL-2	5	OLD

Textbooks:	
1	Discrete and Combinatorial mathematics – Grimaldi and Ramanna – Pearson Education-2017.
2	Discrete Mathematical Structures- Dr. KSC- Sudha Publication-2019.

CO-PO/PSO Mapping

Course Outcome	Blooms Taxonomy Levels	Program Outcomes (PO's) Mapped											
		P01	P02	P03	P04	P05	P06	P07	P08	P09	P010	P011	P012
C01	L3	3	3	-	1	1	2	1	-	2	2	-	-
C02	L4	3	2	-	1	-	-	1	1	1	1	-	-
C03	L2	3	2	-	1	-	-	1	1	1	1	-	-
C04	L4	3	3	1	-	-	1	1	1	-	-	-	-

C05	L3	3	3	-	2	1	2	1	-	2	2	-	-
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